

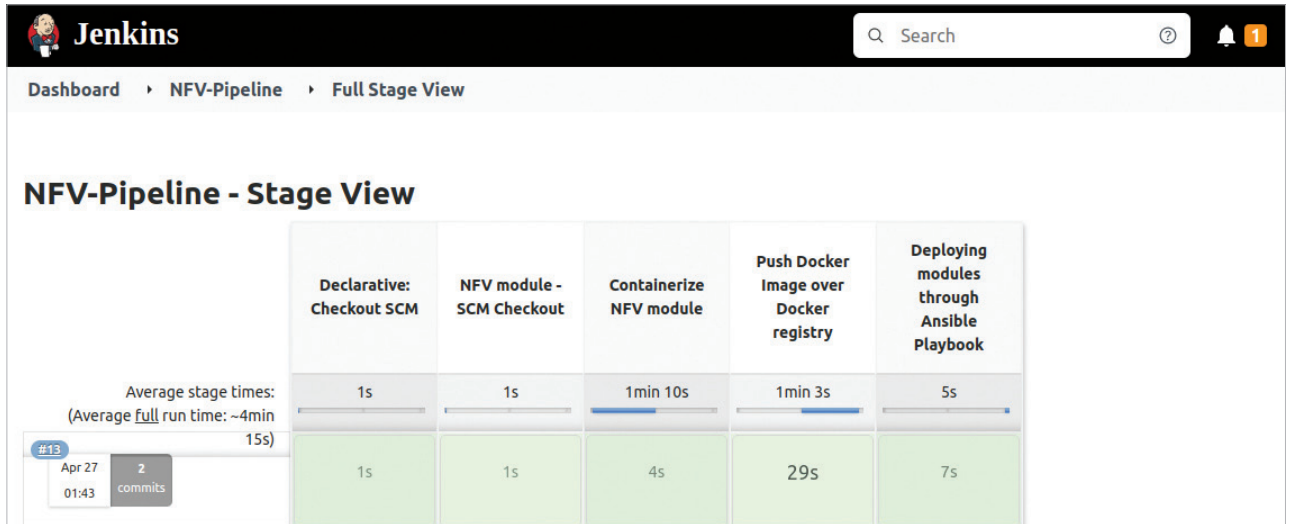


Figure 1: Jenkins pipeline for continuous integration for build and deployment

The NFV application has been created to demonstrate the development and deployment of the NFV module using DevOps tools and methodologies. While the application is purely for experimentation purposes, it can be extended to meet specific customer needs. A VOD service is created and its relevant configuration files and source code can be found at <https://github.com/shubhamaggarwal890/nginx-vod>.

Integrating with Jenkins

Merely creating the code is not enough; developers must build it first to publish it on the production infrastructure. Jenkins, a well-known platform for continuous integration, enables us to achieve precisely that. We use Jenkins with a GitHub webhook, which triggers a notification to Jenkins communicating the latest changes pushed to the GitSCM. Jenkins then pulls the latest changes from GitHub and triggers the created pipeline, where the application is containerised in a Docker container.

This functionality is enabled via the pipeline plugin of Jenkins. All the credentials and environment variables can be stored within Jenkins. Similarly, for this research, a pipeline script has been written to invoke the various stages. The pipeline source code is

given in Figure 1, where we can see that in the *NFV module - SCM Checkout* stage, the git plugin within Jenkins clones the source code repository. Then in the *Containerize NFV module* stage, a Docker image is built on the given source code. The built instructions are forwarded through a Docker file by the installed Docker plugin in Jenkins. Once this stage is completed, the *Push Docker Image over Docker registry* stage is invoked, where the Docker image is pushed to the Docker hub. Finally, the *Deploying modules through the Ansible Playbook* stage calls the Ansible to perform allocated tasks on the designated nodes.

Figure 1 shows the various stages of the pipeline script written to be executed by Jenkins. This script can be found at <https://github.com/shubhamaggarwal890/nginx-vod/blob/master/Jenkinsfile>.

The Git repository is pulled to the Jenkins workspaces, followed by containerisation of the NFV module. After the successful creation of the NFV image, the Docker image is pushed to the Docker registry, i.e., Docker hub. Alongside this information, other details such as the time taken by each stage and the average run time taken by multiple builds are also visible.

Containerisation with Docker

Containerisation of an application has become a norm as well as a healthy habit due to the scalability and resource efficiency provided by containerisation tools. Network functions virtualization (NFV) is also ruled by the microservices. Therefore, the idea here is to containerise different services of the NFV module and connect them via a container orchestrator.

The services are containerised in the Jenkins pipeline and pushed to the Docker registry for the Docker image to be available for the production environment. The containerisation is handled by the Docker plugin in Jenkins. The Docker plugin looks for the Dockerfile within the source code and builds the Docker image from it. The Dockerfile is at <https://github.com/shubhamaggarwal890/nginx-vod/blob/master/Dockerfile>. It invokes the Debian base image and installs the essential packages required to facilitate an environment for the VOD application.

After the successful build of the application, the Docker image is pushed to the Docker hub registry; the image link can be found at <https://hub.docker.com/r/shubhamaggarwal890/nginx-vod>.

Deploying with Ansible

Deployment is often a challenging task as it requires accommodating